

MODULE 2:

1. A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces
 - a. 1000 frames per second?
 - b. 500 frames per second?
 - c. 250 frames per second?
2. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces
 - a. 1000 frames per second.
 - b. 500 frames per second.
 - c. 250 frames per second.
3. Explain Pure ALOHA with procedure
4. With a neat diagram explain CSMA/CD
5. With a neat diagram explain CSMA/CA
6. Mention the different controlled access methods and explain any one in detail.
7. What is the difference between ALOHA and Slotted ALOHA
8. What is PPP? What are the services provided by PPP?

MODULE 3:

1. Discuss the services provided by Network Layer.
2. Explain packet switching datagram approach.
3. Explain Virtual circuit approach to route the packets in packet switched network.(10M)
4. Explain classfull addressing in detail.
5. Explain classless addressing in detail.
6. An organization is granted a block of addresses with beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses in its three subnets.
 - a. One subblock with 10 addresses
 - b. One subblock with 60 addresses
 - c. One subblock with 120 addresses. Design the subblocks.
7. Explain DHCP message format, FSM and operation.

8. An ISP is granted the block 16.12.64.0/20. The ISP needs to allocate addresses for 8 organizations, each with 256 addresses. a. Find the number and range of addresses in the ISP block. b. Find the range of addresses for each organization and the range of unallocated addresses. c. Show the outline of the address distribution and the forwarding table.
9. Explain the IPv6 datagram.
10. Compare IPv4 and IPv6
11. With a diagram explain Least Cost Routing
12. Explain the distance vector routing algorithm with an example.
13. Explain the Link state routing protocol with an example.
14. Explain the path vector routing.
15. Explain the Operation of External BGP (eBGP)
16. Explain the Operation of Internal BGP (iBGP)
17. Discuss the RIP message format
18. Discuss the performance of OSPF
19. Write Dijkstra's algorithm to compute shortest path through graph. (06 M)
20. Write a note on Routing Information Protocol (RIP) algorithm. (04 M)
21. Explain classful addressing system with a neat diagram.
22. Why is subnetting used and explain its importance.
23. Explain Open Shortest Path First Protocol with example.
24. Illustrate the working of OSPF and BGP